Sweet Sweet Revenge

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1. Research Question:

For this project, we were particularly interested in the topic of revenge games in the NBA. With sports media outlets constantly searching for stories, and player movement being at an all-time high, our research question focuses on proving or disproving the narrative that the players play with more emotion during revenge games–exemplified by statistical differences in "revenge games" (games played against a player's former team). We aimed to uncover if NBA players perform significantly differently in revenge games compared to similar non-revenge games, specifically players who were with their previous team for 3 or more years, as these players are more likely to have stronger ties and stronger motivations for revenge performances. Though this has been previously studied, they have mostly focused on older data, often centered around Lebron James' career. With recent blockbuster trades of NBA stars, we will use recent data to capture today's landscape and compare revenge game performances to season averages against teams of similar caliber. We will do so by applying matching techniques based on factors like win totals, team strength, and strength of schedule (SoS).

2. Data Collection and Preparation

One dataset we used for our final project is Basketball Reference, which can be accessed online at https://www.basketball-reference.com/. With this dataset, we primarily focused on understanding and tracking recent player movement. Basketball Reference includes complete NBA data dating back to the 1946–47 season, but for consistency throughout this project, we will restrict our analysis to the 2002–03 season onward. The units of observation were season-level and player-level statistics, allowing us to establish season averages for each traded player. These averages served as a benchmark for comparing performance in revenge games. The key variables include the player name, team, opponent, location of game played, and performance stats such as points, rebounds, assists, plus-minus, and more. Another dataset we used for our final project was the HoopR package from RStudio, specifically the *espn_nba_player_box* tables from the 2002-2003 season onward. For this dataset, the unit of observation is each individual player's performance in a single game. For the season 2023-2024, the dataset has 35,028 rows and 57 total columns, but 15 relevant columns. The key variables for this dataset are similar to the HoopR package. This data source was primarily used for statistical analysis and testing. To clean the dataset, we dropped missing values using drop_na() and formatted data to be consistent throughout both data sources using function clean_names() from R's *janitor* library.

3. Exploratory Data Analysis

athlete_display_name <chr>	games_played <int>	avg_minutes <dbl>	avg_points <dbl>	avg_assists <dbl>	avg_rebounds <dbl>	avg_steals <dbl>
Joel Embiid	50	34.717391	34.2173913	5.60869565	11.0869565	1.17391304
Luka Doncic	93	38.139785	32.3870968	9.36559140	9.2688172	1.50537634
Giannis Antetokounmpo	74	35.081081	30.3378378	6.44594595	11.4594595	1.17567568
Shai Gilgeous-Alexander	86	34.662791	30.0813953	6.22093023	5.7093023	1.90697674
Jalen Brunson	96	35.868132	29.0659341	6.82417582	3.5714286	0.89010989
Kevin Durant	81	37.287500	26.9625000	4.95000000	6.5750000	0.91250000
Devin Booker	73	36.041096	26.9315068	6.89041096	4.4520548	0.95890411
Donovan Mitchell	69	35.469697	26.7878788	5.78787879	5.1666667	1.74242424
De'Aaron Fox	76	36.052632	26.6447368	5.64473684	4.6184211	2.03947368
Nikola Jokic	92	35.239130	26.5434783	8.92391304	12.4021739	1.38043478

1–10 of 589 rows | 1–7 of 10 columns

Previous 1 2 3 4 5 6 ... 59 Next

Figure 1: Individual player summary statistics for the 2023-2024 NBA season

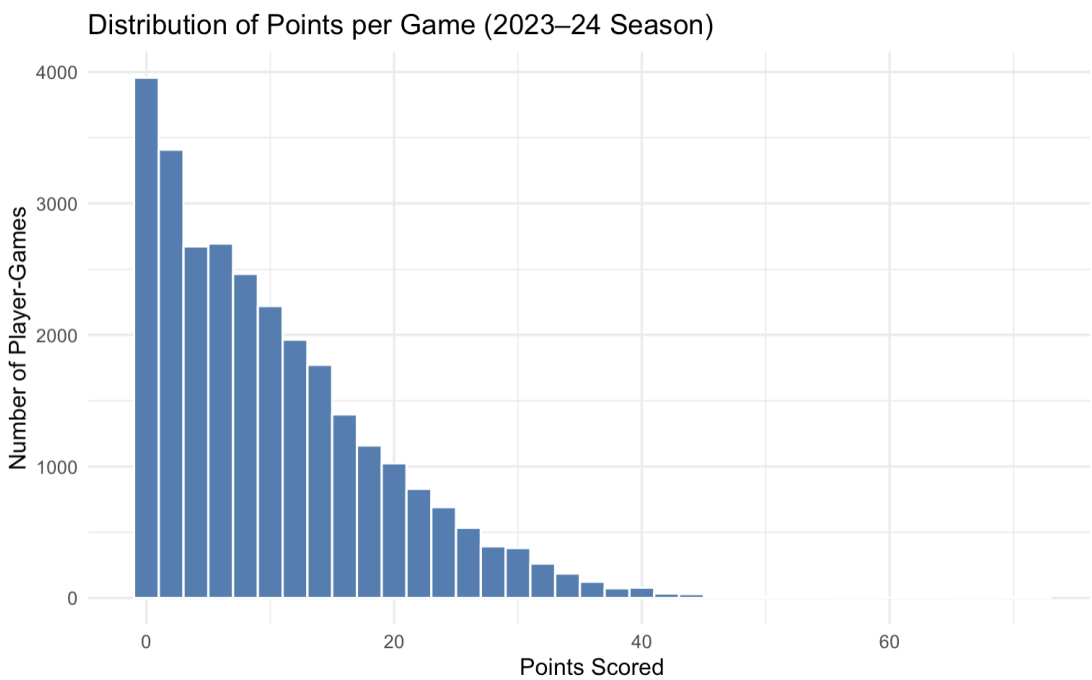

Figure 2: Distribution of the amount of games for points scored per game in the 2023-2024 season

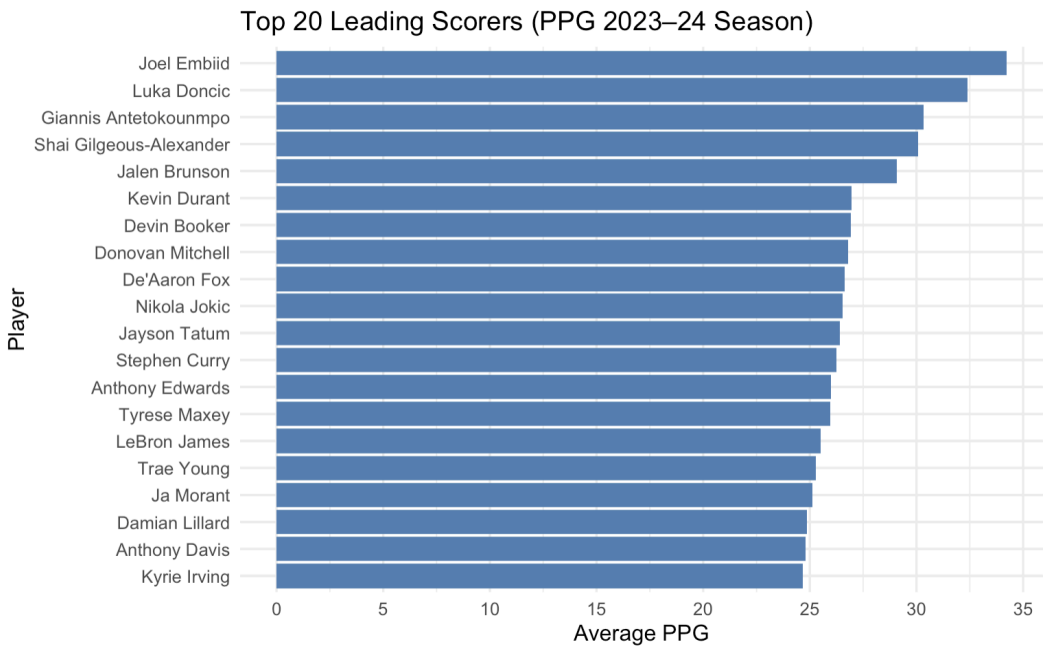

Figure 3: The top 20 leading scorers in the 2023-2024 NBA season

Figure 1 provides information on season statistics, which we can later reference to compare "revenge" games to as a control. Figure 2 helps to provide general information on the type of information we can obtain from the datasets. From Figure 3, we can analyze specific players who have made a bigger impact on the NBA in recent years. From this exploratory analysis, we learned that the HoopR dataset provides clean and structured game-level data ideal for identifying "revenge games." This EDA confirms that using player-level box scores from HoopR will allow us to track individual performance trends before and after trades. We can then filter for "revenge games" of traded players and compare them to each player's season average metrics to test whether players statistically overperform or underperform against their former teams. By keeping these averages for every traded player, we can later match each revenge game's performance against their season average using a one-sample t-test. Additionally, we can match "revenge" games to specific "non-revenge" games with similar opponent strength, rest, expected minutes, etc., which we can then match using the 'MatchIt' package in R.

4. Methodology

In our methodology we started out by looking at a few different approaches when analyzing NBA players and their performance in revenge games. Initially we looked at individual players and specifically pulled stats on the average number of points an individual player scored in a normal game (non- revenge game) and a revenge game. Revenge game being determined by if the game being played was played against their former team in which they were previously on. We charted this information using a bar chart, the x- axis showing the player and

the y-axis showing average points. We sampled 12 players initially just to see if there was a trend amongst individual players and their point performance in the two game types.

In the end we ended up applying the k-nearest neighbors matching method using an R package called Matchit, to create a set of non-revenge "control" games statistically similar to each revenge game. The matching model estimates the probability of a revenge game based on the covariates above and pairs each revenge game with one comparable non-revenge game. After matching, we computed the difference in performance between revenge and matched control games for each statistic: points difference, rebounds difference, assists difference. These values summarize how the player's performance changes specifically in revenge settings, while holding minutes, usage, location, and opponent strength constant. To isolate the true impact of revenge games, we first tagged every game where the player faced their former team. We then used nearest-neighbor matching (via the MatchIt package) to pair each revenge game with a comparable non-revenge game. The matching algorithm controls for minutes played, shot attempts, home-court advantage, and opponent win percentage. After matching, we compute the difference in points, rebounds, and assists between the revenge game and its paired control game. These differences we visualized in bar graphs. We did this for multiple players creating a csv file for each player that we analyzed. The file contains the following columns: player, which includes the player's full name (for individual player csv files the player name was consistent); game_date, which was the date of the matched game that was played; points_diff, rebounds_diff, and assist_diff. We did this for a total of 20 players and in total had 55 games to use for our analysis. We combined each player csv into one csv, having one data set to pull data from so we could effectively run a T-test to see if the difference in means is significant between revenge and matched non-revenge. We also calculated the confidence intervals and p-value for point differences per player, assist differences per player, and rebound difference per player. Creating three scatterplots for each of these outcome variables. The figures display a per-player comparison of the specified outcome variable performance in revenge games versus matched non-revenge games. Each point or dot represents the difference for an individual revenge game. A horizontal dashed line at 0 served as a reference with points above the line reflecting improved performance, while points below reflect a decline.

5. Results

Our analysis focused on whether traded NBA players really do play significantly differently against their former teams. By using nearest-neighbor matching techniques, we aim to see if games played against their former teams vary against games played against similar opponents. To contextualize this, we can ask: "Does Luka Doncic play significantly differently against the Mavericks?" To do this, we see if stats against teams of similar strength to the Mavericks are statistically significantly different to when he plays the Mavericks.

After collecting revenge game data for about 20 players in the NBA's Top 75 list (including Luka Doncic, Karl-Anthony Towns, LeBron James, etc), we created a pool of 56 revenge games and matched-game pairs. Each pair of games compared a player's performance in

a revenge game to a player's performance in a matched game of similar caliber using R's MatchIt package (using default nearest-neighbor matching). We used matching statistics using revenge game status (marked by a custom-made variable), matched on minutes-played, field-goal attempts, home/away status, and opponent win percentage.

5.1 Summary of Revenge Effects

When analyzing the players, the average differences between revenge and matched-game performance were minimal:

- +0.93 points (indicating a higher scoring role, likely due to personal "revenge")
- -0.2 rebounds (interesting find, as most interviews show that players want to fight more for rebounds)
- -0.46 assists (wanting to hold the ball more - prioritizing scoring)

These values indicate that, on average, players perform virtually identically in revenge games compared to their matched non-revenge counterparts. It's important to note, however, that these metrics are the **averages** of **all players**, meaning an assortment of players can and do perform significantly differently in their revenge games compared to matched games.

**Most important limitation to this study is the relatively small sample size, N=55*

5.2 Formal Statistical Tests

After computing the differences between the control and experimental groups, we had to find if there was statistical significance in our results. To do this, we relied on a series of one-sample t-tests on the difference of revenge game versus matched game statistics.

Our results found no statistically significant differences:

Statistic	*Mean Difference*	*T-value (t)*	*N (number of games)*	*p-value*	*95% Confidence Interval*
Points	0.93	0.512	55	0.6108	[-2.71, 4.56]
Rebounds	-0.20	-0.357	55	0.7225	[-1.299, 0.906]
Assists	-0.46	-1.037	55	0.3045	[-1.362, 0.433]

Table 1: One-Sample t-test Results

As seen in Figure 1, all confidence intervals include 0, and all p-values exceed 0.05 by a sizable margin. This suggests that the observed differences are too small to be considered significance and are indistinguishable from random noise from the data collected.

The t-values, which measure the statistical significance of the difference in means, are too low to note any real difference in points, rebounds, and assists. T-values close to 0 are considered too small, as it indicates the difference less than 1 standard deviation from the mean. However, we see that *t* for assists is far different from the rest. Although this value still does not align with the critical value associated (±1.96), it shows that further research (with a higher sample size, *N*) may show there is something more than just noise.

5.3 Visualizing the Revenge Effect

We primarily used a scatter plot to visualize a player's difference in a revenge game, using the matched games as a baseline (of 0) to quantify how much better (or worse) they played in revenge games.

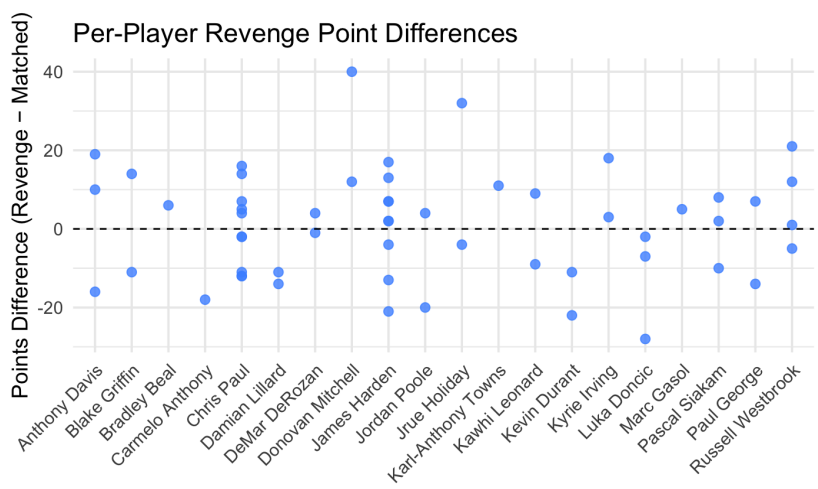

Figure 4: Per-Player Revenge Point Differences

As seen in Figure 4, there does not exist a clear trend in point differences in revenge and matched games. Using baseline-offset as our comparison metric, the data indicates that a select few players really do play significantly different from matched games. These players include Donovan Mitchell, Anthony Davis, Kevin Durant, Russell Westbrook, Jrue Holiday, and Kyrie Irving. This plot ultimately shows that the distribution is centered around zero, with an almost symmetric variation in both directions. Although the aforementioned players show clear

differences in revenge games, these performances are counterbalanced by underwhelming performances.

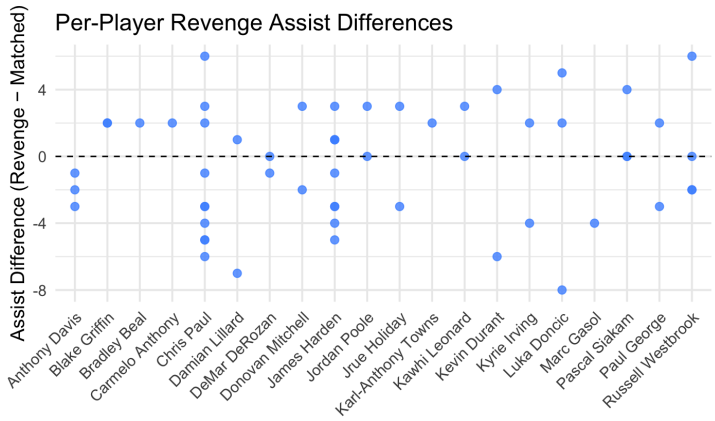

Figure 5: Per-Player Assist Differences

As shown in Figure 5, it's apparent there's slightly more clustering below the baseline. This shows that players are holding onto the ball more–indicative of taking on a higher scoring role. Despite this, there does not exist a clear trend, as symmetry across both sides of the chart exists. As previously mentioned, further study with a higher sample size of games could prove that there is statistical significance in players having less assists in revenge games. As for our study, statistical tests showed that there does not exist a true correlation between the two variables.

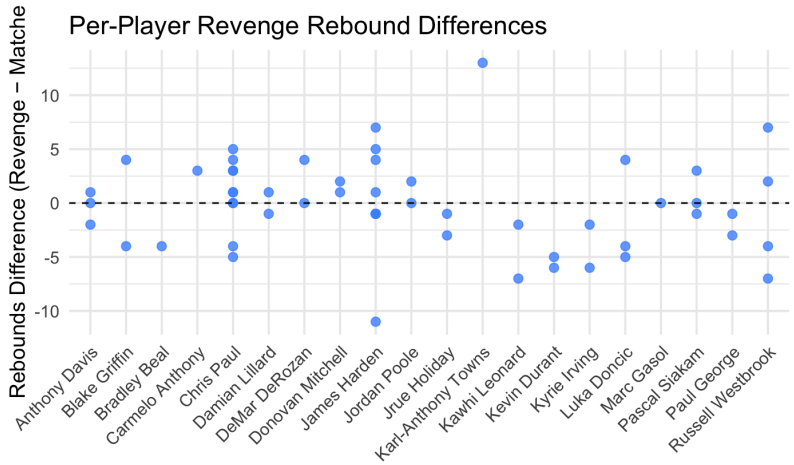

Figure 6: Per-Player Revenge Rebound Differences

Finally, Figure 6 presents an interesting find. As opposed to popular belief, players actually seemed to have **less** rebounds in their revenge games. Normally, pundits and fans alike

believed that players would, and should, fight harder for rebounds (offensive *and* defensive) to show more physical dominance against their former teammates. However, Figure 6 and its corresponding difference in mean instead indicate that players have *less* rebounds in revenge games. Even dedicated bigs and centers (Anthony Davis, Blake Griffin, etc) show revenge stats close to the baseline of 0.

5.4 Interpretation

All-in-all, despite the ongoing media narratives around emotional motivation and player "revenge arcs," our analysis shows that, once contextual factors are controlled, players perform very to their baseline expectations. Individual variation exists between select players, but when aggregating, revenge games do not produce significant performance changes.

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6. Discussions and Conclusions

Our research set out to answer whether NBA players perform significantly differently in revenge games compared to statistically similar non-revenge games. After applying nearest-neighbor matching and running T-tests across points, rebounds, and assists, we found no statistically significant differences in player performance. These results indicate that, on average, facing a former team does not lead to a measurable boost or decline in performance. While select players displayed small individual tendencies, these were not strong or consistent enough to influence league-wide trends.

Therefore, our findings do not support the popular narrative that revenge games inherently motivate players to perform better. Instead, the data suggests that performance is largely stable regardless of opponent, and emotional storylines do not translate into significant statistical outcomes. However, our study is limited by sample sizes for certain players and by the variables available for matching. Future work could expand to additional performance metrics, incorporate play-by-play efficiency measures, or analyze psychological and contextual factors that are harder to quantify.

In conclusion, revenge games may create compelling narratives, but they do not meaningfully change on-court performance across the NBA, providing a clear, data-driven answer to our research question.